Kuldip Patel

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Education

National Forensic Sciences University **July 2025 – Aug 2027 (Expected)**
Master of Technology in AI & Data Science, GPA: 8.15/10.00 *Gandhinagar, Gujarat, India*

Completed Coursework: Mobile Security and Forensics, Network Security and Forensics, Advanced Machine Learning, Natural Language Processing, Deep and Reinforcement Learning, Intelligent Systems and Security

San Jose State University **Jan 2022 – Dec 2022**
Graduate Coursework in Data Analytics, GPA: 3.33/4.00 *San Jose, California, USA*

Completed Coursework: Big Data Technology and Applications, Database Systems for Analytics, Machine Learning

Gujarat Technological University **July 2016 – Aug 2020**
Bachelor of Engineering in Computer Engineering, GPA: 7.99/10.00 *Surat, Gujarat, India*

Projects

YALM Model | *Python, Pytorch, Transformers, W&B* **July 2025 – Sep 2025**

- Trained 80M and 130M parameter decoder-only transformer language models on a 62M-row (∼**42B** tokens) multilingual dataset (English, Hindi, Code, Math).
- Built training and evaluation pipeline, custom BPE tokenizer (**32K** vocab), and dataset preprocessing workflows.
- Reduced training time by **50%** by scaling training on multi-GPU using DDP method with BF16 precision.
- Evaluated models on standard benchmarks achieving competitive scores for sub-200M parameter models (**27%** MMLU, **33%** ARC).

GPT Trainer | *Python, Pytorch, Transformers* **May 2025 – July 2025**

- Built a modular GPT-style training framework from scratch to experiment with architecture and optimizer variations.
- Implemented core transformer components: causal multi-head self-attention, feed-forward layers, layer normalization, and sinusoidal/rotary (RoPE) positional embeddings.
- Trained 20M parameter Llama-style Small LM for ablation studies of the YALM project, enabling **20+** experiment variations.

IDA Github Actions | *Python, Linux, Github Actions* **Nov 2024 – Dec 2024**

- Built a CI pipeline using GitHub Actions to automate static binary analysis with Interactive Disassembler (IDA).
- Processed 3 analysis tasks by developing Bash and Python scripts to configure environments, orchestrate execution, and collect artifacts.

Corona Symptoms Detection Chatbot | *Python, Flask, Spark, Socket.io* **Apr 2022 – May 2022**

- Built real-time Covid-19 symptom detection chatbot with Flask backend, JavaScript frontend, and Socket.io for bidirectional communication, supporting **5+** concurrent users.
- Achieved **87%** classification accuracy by training a Random Forest classifier on a large-scale dataset using PySpark.

Restaurant Risk Prediction | *Python, Pandas, Matplotlib, Scikit-learn* **Nov 2022 – Dec 2022**

- Implemented capstone project for Machine Learning subject regarding prediction of restaurant risk category.
- Collaborated with team of 4 in preprocessing, analysis and visualization of dataset also creation and comparision of different ML models, achieved 97% accuracy on dataset.

Technical Skills

Programming: Python, SQL, Golang, Java, JavaScript, PHP, Bash, C/C++
Security: Nmap, Wireshark, IDA Pro, Ghidra, Autopsy
AI/ML: PyTorch, Hugging Face Transformers, NumPy, Tokenizers, LLM Fine-tuning
Tools/Platforms: Docker, AWS, Git/Github, MySQL, MongoDB, JupyterLab, Flask, Node.js, ReactJS

Experience

Freelance Software Developer **Sep 2020 – Jun 2025**
Self-Employed *Remote*

- Boosted client productivity **40%** by building and maintaining web application using React, Node.js and MySQL, streamlining client workflows.
- Cut manual data collection by **10** hours per week by creating data scraping and automation pipelines using Python.
- Increased lead capture rate by **15%** and served **1,200+** daily visitors through designing and maintaining a Bootstrap landing page.